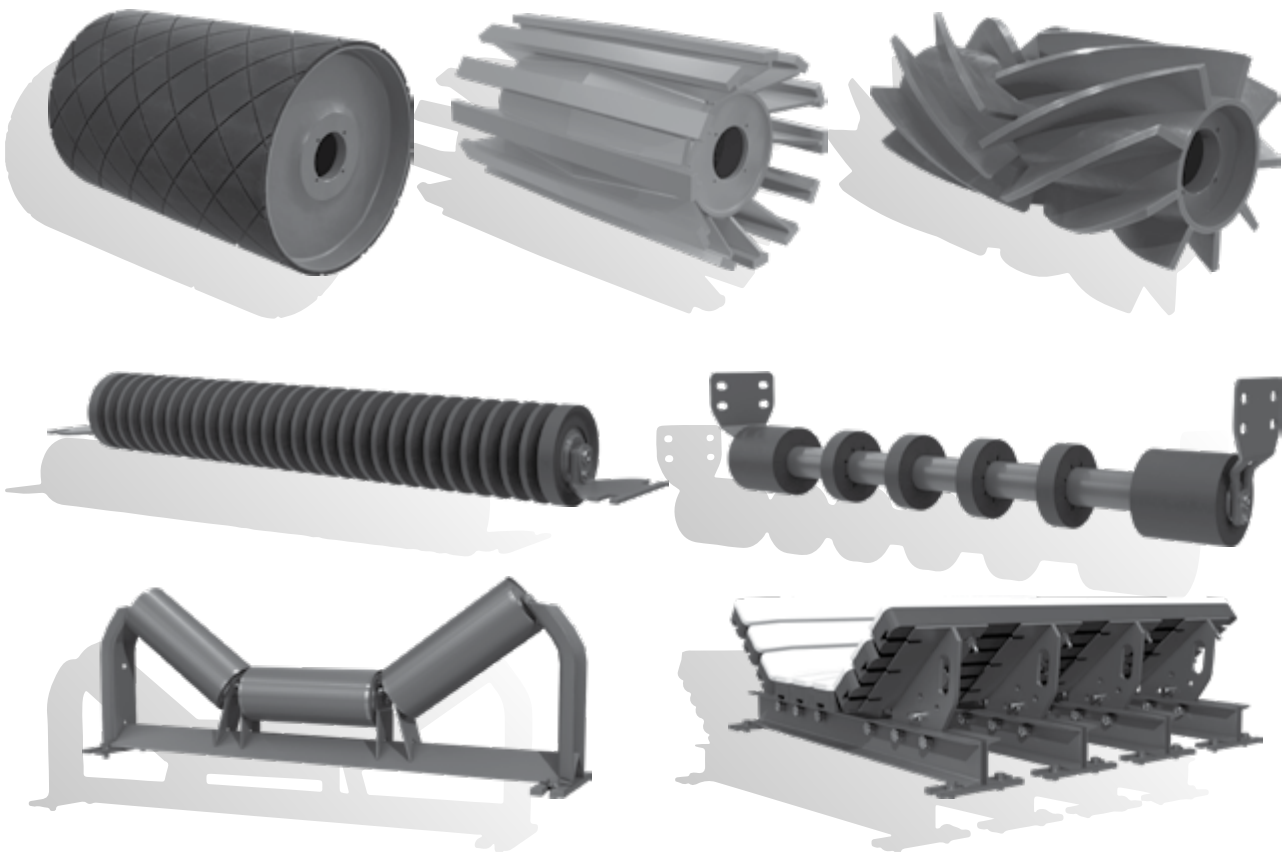
Martin



PULLEYS AND IDLERS

INSTALLATION, OPERATION AND MAINTENANCE MANUAL

INSTALLATION
OPERATION
MAINTENANCE
INSTRUCTIONS

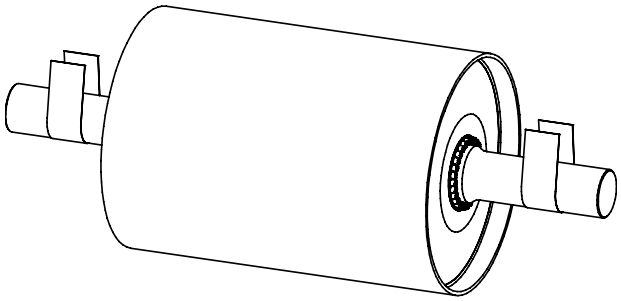
CONVEYOR PULLEY INSTALLATION

Installation Instructions:

1. Failure to follow the installation and operation instructions outlined below could result in serious injury. Follow all safety guidelines that pertain to your company's lockout/tagout/blockout procedures before performing maintenance or installation.
2. Remove debris from underside of the belt prior to installation. Failure to do so can result in belt damage and/or premature pulley wear.
3. Use appropriate lifting equipment when handling the pulley for assembly and installation. Never lift the pulley using the bearing housing.

   Lifting hardware on the bearing is not designed to support the pulley assembly weight.

4. Place weight rated nylon lifting straps around the pulley shaft between the pulley face and bearings. Place the straps as close to the pulley face as possible. Never use a chain to move a conveyor pulley assembly.



Alternate
lifting diagram
for pulley assemblies

5. Pulleys delivered on skids or cradles can be lifted with fork trucks, if trucks are rated for the weight capacity and forks extend through the skid or cradle.
6. Ensure pulleys are mounted square and perpendicular to the belt to help ensure best tracking. Tighten bearing base bolts securely and to bolt size and grade specifications (not included).
7. If taconite seals are used, ensure pulley misalignment from bearing to bearing is limited to 0.5 degrees to eliminate component binding and heat buildup or premature wear.

Assembly Procedure:

1. Identify all components for assembly and any specific assembly instructions or component requirements.
2. Clean all components and make sure shafts and bore of pulley hub and locking devices are free of dirt or contamination. Remove any protective coating that may be on shaft surfaces.
3. Place and secure pulley on secure platform or bench for assembly.
4. Install locking device onto the shafts and into the hubs of the pulley by following manufacturer's instructions. Do not apply lubricant to the hardware when installing as this can lead to premature failure.
5. Refer to locking device instruction manual for tightening and torque requirements. Use calibrated torque wrenches to tighten hardware loosely and ensure shaft is in the proper position.
6. Accurately torque the locking hardware per manufacturer's instructions. Confirm pulley is in correct position and properly aligned, as the axial position can move during tightening of locking hardware.
7. Check mounting surfaces and holes before mounting bearings and housing. Follow manufacturer's instructions for proper installation and alignment to ensure bearing and seal life. Secure bearings to the shaft and tighten the fixed/non-expansion bearing before tightening the floating/expansion bearing.
8. If coupling hubs are required, follow manufacturer's installation manual. Verify all coupling components are placed onto shaft before coupling hub. Install hubs based on clearance or interference fit instructions.
9. If any other components are required, follow manufacturer's instruction manual.
10. If not placed into service immediately, protect the entire pulley assembly from direct sunlight, rain, snow, or larger temperature and/or humidity variations.

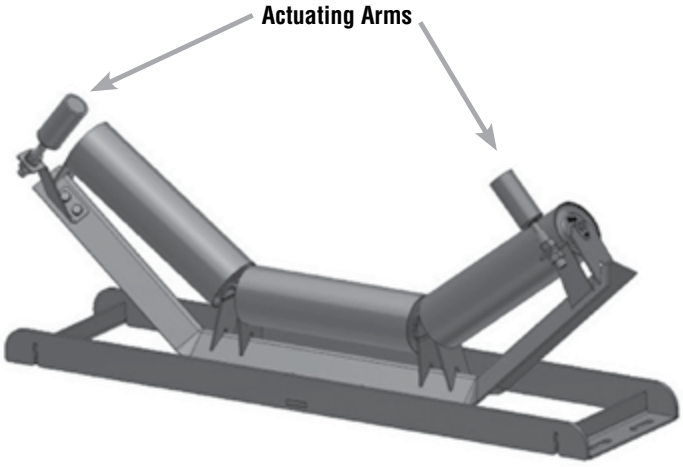
General Operation Instructions:

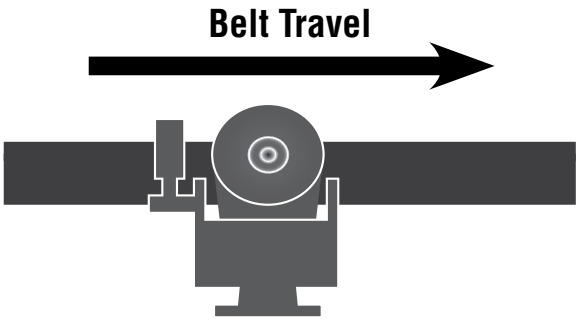
1. Inspect and torque locking device weekly in the first month of operation. If bolts are loosening, review pulley load and shaft deflection. Excessive shaft deflection may cause bolt retention issues. Do not over torque bolts.
2. Refer to the bearing manufacturers installation, operation and maintenance schedule for all bearing related instructions.
3. Inspect pulley periodically for lagging wear, cracks and rim adhesion. Deterioration may occur if oil, grease, kerosene, solvents or other chemicals remain on lagging.
4. Inspect pulley end discs and shell for cracks and signs of stress fatigue. Never operate a pulley with a cracked end disc and shell. Never drill, weld or make modifications to pulley end disc and shell as this may lead to pulley failure.
5. Inspect all take-up, snub and belt tensioning devices to ensure proper function and movement during operation. Improper belt tensions may result in component failure or belt and pulley wear due to excessive slipping.
6. Monitor bearings for proper alignment, movement and heat during operation. Use a dial indicator, check both elevation and squareness to the shaft is within manufacturer's recommended limits. If excessive movement occurs during operation, the bearing may not be properly secured to the shaft or bearing wear has occurred. A rise in temperature could indicate bearing failure. Re-lubricate per manufacture's recommendation.
7. Modification, repair or other work on a conveyor pulley assembly should not be performed without prior written consent of Martin Sprocket & Gear.

IDLER INSTALLATION

CAUTION: ALWAYS LIFT IDLERS BY THE FRAME, NEVER BY THE ROLLS.

1. Establish the conveyor centerline by use of a wire or string.
2. Remove mud, stones, burrs, or any other debris from the stringers so that the Pulleys and Idlers will sit squarely in position. This precaution will help prevent belt training problems.
3. Mount Pulleys level with the plane of the conveyor and perpendicular to the line of belt travel. Do not adjust the Pulleys after they have been accurately positioned.
4. Position Idlers perpendicular to the line of belt travel. **THE MIDDLE OF EACH CENTER ROLL MUST BE ON THE CENTERLINE OF THE CONVEYOR**. Tighten all four mounting bolts securely.
5. Rotate each roll to be sure it turns freely. If a roll is tight, look for some external interference or evidence of damage to roll or frame.
6. Install training Idlers with the same care and accuracy used in mounting the basic carrying and return Idlers.
7. To prevent damage during shipment, the actuating arms of positive action training Idlers are **NOT** mounted in operation position at the factory. Bolt the arms to the swivel frame so they extend **TOWARD THE APPROACH SIDE OF BELT**.





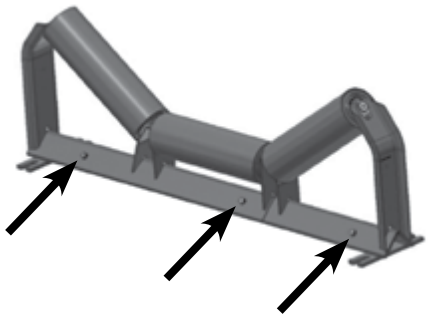
8. Inspect roll surfaces and remove any foreign material, especially abrasive dust, to prevent damage to the underside of the belt.
9. Install belt.

NOTE: All Martin Idlers are permanently lubricated at the factory before shipment.

Retractable Idlers:

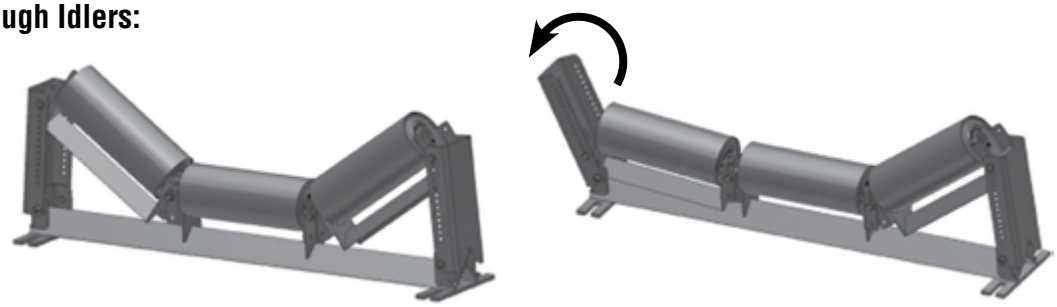


1. When working with Retractable Idlers, always follow operation instructions and ensure conveyor belt is stopped and conveyor is powered down with lock out/tag out safety features. Recommendation would be to ensure belt is empty prior to any work on idlers.
2. Loosen and remove bolts in frame base.

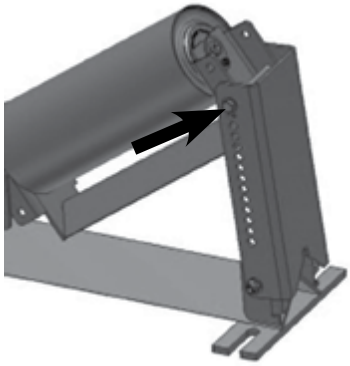


3. Frame can then be pulled apart from each side of the belt for roll removal or inspection.
4. Reverse steps to re-insert idler frame back into place, align holes in frame and secure all bolts back to recommended torque specs (70 ft-lbs).
5. Conveyor belt can then be restarted once all components are securely fastened and personnel are safely away from conveyor belt.

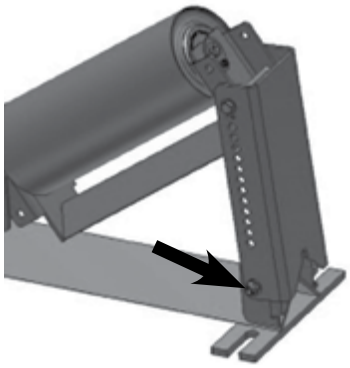
Adjustable Trough Idlers:



1. Standard Adjustable Idlers can be used to change belt trough from 5 to 35 degrees in 2 ½ degree increments. Optionally, an adjustable idler can be provided with 45 degree max trough.
2. When adjusting idler trough angle, ensure conveyor belt is stopped and conveyor is powered down with lock out/tag out safety features. If belt is still full of material, ensure it can not spill on operators when making idler adjustments. Recommendation would be to ensure belt is empty prior to any work on idlers.
3. Loosen bolts on each side of end bracket that attach to bracket on idler wing roll. When removing bolts be sure to support angled wing rolls prior to removing bolt to avoid it dropping suddenly.

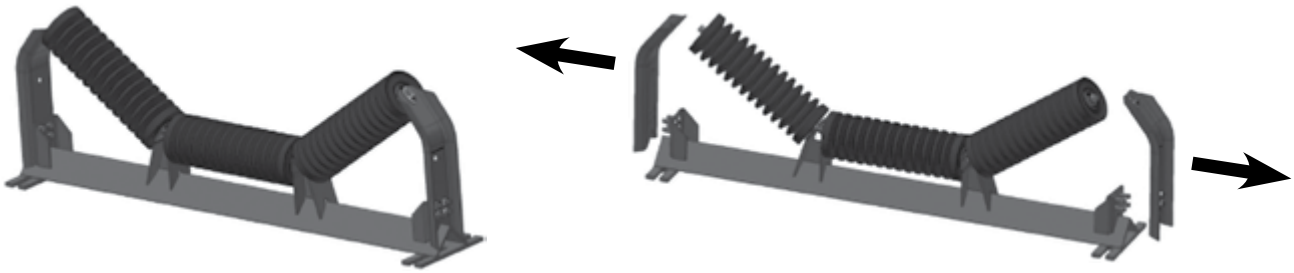


4. Loosen nut at bottom of end bracket to allow bracket to rotate and align with bracket on wing roll.

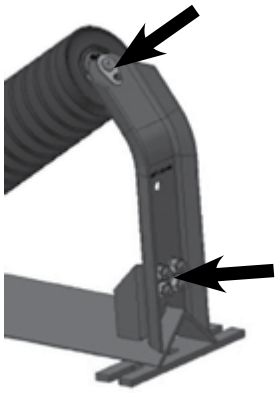


5. Wing roll and end bracket are now free to rotate and change trough angle as needed.
6. Once required angle is achieved, re-insert bolts on end bracket back into the roll end bracket to secure in place, tighten nut at bottom of end bracket and torque all bolts/nuts back down securely.
7. Conveyor belt can then be restarted once all components are securely fastened and personnel are safely away from conveyor belt.

Removable End Bracket Idlers:



1. When working with Removable End Bracket Idlers, always follow operation instructions and ensure conveyor belt is stopped and conveyor is powered down with lock out/tag out safety features. Recommendation would be to ensure belt is empty prior to any work on idlers.
2. Loosen and remove bolts at bottom of end brackets and any clips holding idler shaft to end bracket.



3. End Bracket can now be removed out the side of the idler. Support wing rolls when removing end brackets to ensure roll does not fall out of frame.
4. Remove any center clips holding rolls to center brackets.



5. Rolls can now be removed from frame.
6. After rolls have been replaced back into frame, re-insert all center clips and use bolts and nuts to re-attach end brackets to the frame securely. Once bracket is back in place end clips can be re-installed to wing roll shafts.

CAUTION: Before performing any maintenance, the circuit should be opened at the switch box, and the switch should be padlocked in the **OFF** position.

After the Idlers and belt have been installed, the system should be started while empty and checked for alignment. A properly aligned conveyor has the belt running evenly in the center of the Idlers and as a result, prevents injury to the belt edges from contact with supporting structures or other objects. If a misalignment problem exists, it is not advisable to attempt correction by readjusting the head or tail Pulley because undue strains on the Pulleys, bearings, belt, belt splice or joint or the conveyor may result. Pulleys should be carefully aligned when installed and should not be adjusted for purposes of belt training.

1. Check the alignment of the entire system by operating the conveyor with the belt completely empty. If all components are properly aligned, the belt will run evenly in the center of the Idlers.

2. If misalignment exists and the empty belt does not run true, DO NOT attempt to correct the problem by adjusting the head or tail Pulley. This causes undue stress on bearings, belt splices, and conveyor frames without correcting the problem.

   ⚠ **CAUTION**: Failure to follow these precautions may result in serious **PERSONAL** injury or damage to equipment.

   If one section of belt consistently runs out of line, either the belt is not straight or the splice is not square.

   Proper alignment is achieved by loosening the mounting bolts on several Idlers on the upstream side and skewing them slightly. When one side of an Idler is shifted ahead of the other, the belt shifts to the side that is behind. To make adjustments use the following steps:

   Loosen the mounting bolts and shift the run-out side forward until the belt runs true.

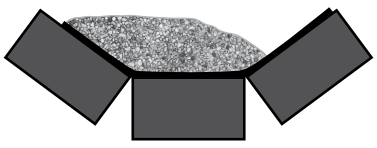
   Retighten mounting bolts, restart conveyor, and check entire system.

   Belt straightness, squareness of splices, and Idler alignment are of vital importance when a belt conveyor operates in both directions.

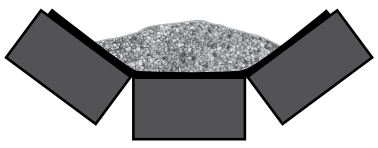
3. Training Idlers are permanently adjusted so that their upper base can swivel 10° in either direction.

   Check belt alignment under full load conditions. A properly aligned and loaded belt will run in the center of the Idlers and will NOT require corrective action from the trainers. This capability is reserved to compensate for occasional off-center belt loading, the effects of wind, concentration of lumps, and other variations which cause temporary belt misalignment.

   Recurrent of continuous misalignment is usually caused by off-center belt loading. Adjust chute so load is distributed as evenly as possible in the center of the belt.



Offset Loading
Affects Belt Alignment



Central Loading
Maintains Belt Alignment

CONVEYOR INSPECTION

Costly interruptions in production can often be avoided by a program of regularly scheduled inspections of the system and all components.

Many operators and maintenance supervisors find it economical to inspect the belt daily for breaks in the rubber covering or signs of edge rubbing. It is better to make this inspection before the system is operated and while the belt is empty.

1. After start-up, check belt loading. Chutes should deliver an even flow of material and load it centrally on the belt.
2. Check for unusual vibrations – they can loosen mounting bolts, allow Idlers to shift and cause misalignment. If this condition occurs, eliminate the cause; then realign all loose Idlers and retighten the mounting bolts.
3. Be sure spilled materials do not interfere with swiveling of training Idlers, or the free rotation of Idler rolls. Good housekeeping is essential to high operating efficiency.
4. If an Idler is sluggish, but its movement is not retarded by material buildup, a choked interior or an impending bearing failure is indicated. The latter condition is almost always signaled by an unusual noise, generally a high-pitched squeal. Sluggish, noisy, or completely stalled rolls require immediate attention because they waste power and cause excessive belt wear. If stalled rolls remain in the system, the outer shell will eventually wear through and the resulting sharp edges will severely damage the belt. When a faulty roll is discovered, tag the Idler immediately and remove it from the conveyor as soon as the system is shut down.

IDLER MAINTENANCE

Lubrication

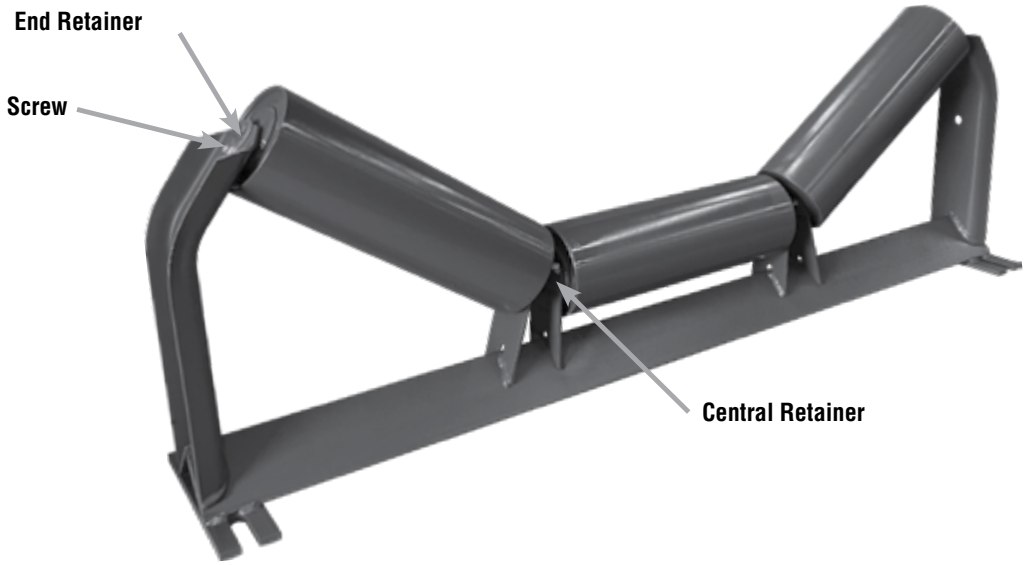
Martin Idlers are factory greased and sealed for life and therefore will not require any additional grease during their life cycle.

For high ambient temperature ranges, consult Martin Sprocket and Gear, Inc. or your lubricant supplier for recommendations.

CAUTION: Before performing any maintenance, the circuit should be opened at the switch box, and the switch should be padlocked in the **OFF** position.

1. Remove mounting bolts.
2. Tip Idler forward or backward, whichever is more convenient, until it rests on decking. If the installation does not include decking, additional precautions must be taken to prevent the Idler from falling through to the return run and causing damage or injury. A suitable plank might be used to support the Idler when it comes to rest.
3. Slide the Idler out from under the belt.

NOTE: When Idler is reinstalled, be sure it is accurately aligned and securely bolted in place.



Roll Removal

1. Remove end and center retainer clips.
2. Remove both end rolls. It may be necessary to tap the rolls to free them from the brackets. Use a rubber head mallet.
3. Lift Idler rolls out of the frame.

Roll Installation

1. Place center roll in Idler frame.
2. Install end rolls and secure with end and center retainers. NOTE: End and center retainers are an integral part of the Idler assembly and must be installed.

Type Of Failure	Probable Cause / Corrective Action
Belt Wear / Breakage	
Excessive Bottom Cover Wear	• **Material build-up (on pulleys and idlers).** Remove the accumulation and install cleaning devices, scrapers, and inverted "V" decking. • **Idlers frozen.** Free the idlers. • **Breaker strip missing or inadequate.** When service is lost, install belt with proper breaker strip. • **Insufficient traction between belt and pulley.** Increase wrap with snub pulleys. Lag drive pulley. In wet conditions use grooved lagging. Install the correct cleaning devices on belt and centrifugal switch for safety. • **Material falling between belt and pulley.** Use skirtboards properly. Remove accumulation. • **Pulley lagging worn.** Replace worn pulley lagging. Use grooved lagging for wet conditions. Repair loose bolts protruding. • **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew.** Use belt designed for specific condition. For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun.
Belt Folding Over On Itself	• **Severe pulley crowning** • **Material buildup** • **Component alignment**
Short Breaks In Carcass Parallel To Belt Edge, Start Breaks In Carcass	• **Impact of material on belt.** Redesign chute so as to reduce impact; install impact idlers, or impact bed. • **Material trapped between belt and pulley.** Install plows or scrapers on return run ahead of tail pulley.
Belt Sag	• **Insufficient belt tension or improper idler placement.** Adjust placement of idlers and adjust belt tension as necessary.
Belt Splices Are Separating	• **Particles from buildup migrate and grind into top cover and in between small imperfections in a belt splice.** Use good quality skirting at loading points and other spots where spillage is likely. Install lagging on the head and snub pulleys. Install a belt plow to prevent trapping material between the belt and tail pulley.
Vulcanized Splice Separation	• **Pulleys too small.** Use larger diameter pulley. • **Drive underbelted.** Recalculate maximum belt tensions and select correct belt. If line is over-extended, consider using two flight system with transfer point. If carcass is not rigid enough for load, install belt with proper flexibility when service is lost. • **Material build-up (on pulleys and idlers).** Remove accumulation and install cleaning devices, scrapers, and inverted "V" decking. • **Excessive tension.** Recalculate tension and use appropriate belt • **Insufficient traction between belt and pulley.** Increase wrap with snub pulleys. Lag drive pulley. In wet conditions use grooved lagging. Install correct cleaning devices on belt and centrifugal switch for safety. • **Belt improperly spliced.** Retighten after running for a short while. If improperly spliced, remove old splice and resplice. Set up regular inspection schedule.
Belt Hardens Or Cracks	• **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew: use belt designed for specific condition.** For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun. • **Pulleys too small.** Use a larger diameter pulley. • **Pulley lagging worn.** Replace worn pulley lagging. Use grooved lagging for wet conditions. Repair loose bolts protruding.

Type Of Failure	Probable Cause / Corrective Action
Cover Swells Or Softens In Spots	• **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew: use belt designed for specific condition.** For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun. Do not over lubricate idlers.
Covers Harden Or Crack	• **Heat or chemical damage.** Use belt designed for specific condition. • **Improper storage or handling.** Follow recommendations for proper storage or handling instructions.
Cover Blisters Or Sand Blisters	• **Cover cuts or very small cover punctures allow fines to work under cover and cut cover away from carcass.** Make spot repair with vulcanizer or self-curing repair material. • **Spilled oil or grease.** Over-lubrication of idlers; improve housekeeping; reduce quantity of grease used; check grease seals.
Ply Separation	• **Insufficient traverse stiffness.** Replace with the proper belt. • **Excessive tension.** Recalculate and adjust tension. Use vulcanized splice with recommended limits. • **Pulleys too small.** Use larger diameter pulleys. • **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew: use belt designed for specific condition.** For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun. • **Excessive impact on belt and splice**. • **Edge worn or broken: (permitting moisture penetration and belt shrinkage on one side).** Repair belt edge. Remove badly worn or out-of-square section and splice in new piece of belt. • **Belt edge contacting structure**. Check for obstructions. • **Belt speed too fast.** Reduce speed.
Belt Breaks At Or Behind Fasteners, Or Fasteners Tear Loose	• **Belt improperly spliced or wrong fasteners.** Use fasteners recommended by the manufacturer. Retighten after running for a short while. If improperly spliced remove old splice and resplice. Set up regular inspection schedule. • **Pulleys too small.** Use a larger diameter pulley. • **Excessive tension.** Recalculate and adjust tension. Use vulcanized splice with recommended limits. • **Pulley lagging worn.** Replace worn pulley lagging. Use grooved lagging for wet conditions. Repair loose bolts protruding. • **Material falling between belt and pulley.** Use skirtboards properly. Remove accumulation. • **Drive underbelted.** Recalculate maximum belt tensions and select correct belt. If line is over-extended, consider using two flight system with transfer point. If carcass is not rigid enough for load, install belt with proper flexibility when service is lost. • **Counterweight too heavy.** Recalculate weight and adjust counterweight accordingly. Reduce take-up tension to point of slippage; retighten slightly. • **Differential speed wrong on dual pulleys.** Make necessary adjustment and observe operation closely.
Excessive Top Cover Wear, Uniform Around Belt	• **Dirty, stuck, or misaligned return rolls.** Remove accumulations; install cleaning devices; use self-cleaning return rolls; improve maintenance and lubrication. • **Cover quality too low.** Replace with belt of heavier cover gauge or higher quality rubber or other elastomer. • **Material spillage and build-up.** Improve loading and transfer conditions; install cleaning devices; improve maintenance. • **Off-center loading or poor loading.** Adjust chute to place load on center of belt; discharge material in direction of belt travel at or near belt speed. • **Excessive sag between idlers causing load to work and shuffle on belt as it passes over idlers.** Increase tension if unnecessarily low; reduce idler spacing.

Type Of Failure	Probable Cause / Corrective Action
Excessive Belt Stretch	• **Tension too high.** Increase speed, same tonnage, same speed; improve maintenance to remove friction and replacement of damaged idlers; decrease tension by increasing arc of contact or go to lagged pulley; reduce CWT to minimum amount. Use vulcanized splice with recommended limits. • **System underbelted.** Recalculate belt tensions and select proper belt. Recalculate maximum belt tensions and select correct belt. If line is over-extended, consider using two flight system with transfer point. If carcass is not rigid enough for load, install belt with proper flexibility when service is lost. • **Build-up of material on idlers.** Remove accumulation; improve maintenance. Install scrapers or other cleaning devices. • **Counterweight too heavy.** Lighten counterweight to value required by calculations. • **Differential speed wrong on dual pulleys.** Make necessary adjustment and observe operation closely. • **Insufficient counterweight travel.** Check for recommended minimum distances. • **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew.** Use belt designed for specific condition. For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun.
Belt Breaks At Or Behind Fasteners: Fasteners Pull Out	• **Fastener plates too long for pulley size.** Replace with smaller fasteners; increase pulley size. • **Wrong type of fastener, fasteners too tight or too loose.** Use proper fastener and splice technique; set up schedule for regular fastener inspection. • **Tension too high for fasteners.** Use vulcanized splice. • **Pulleys too small.** Use larger diameter pulleys. • **Interference from belt scrapers.** Adjust belt scrapers. • **Belt carcass too light.** Select stronger carcass.
Belt's Top Cover And Belt Edges Are Wearing Excessively	• **Build-up on the snub pulley and return idlers often wears the top cover.** Reduce the wear by installing rubber or plastic sleeves on the return idlers and smooth lagging on the snub pulley. • **Improper load can also damage the top cover and belt edges.** • **Off-center loading or poor loading.** Adjust chute to place load on center of belt; discharge material in direction of belt travel at or near belt speed. • **Create a stable, positive seal between the chute's bottom edge** (or skirtboard) and the belt by using impact rollers or slider beds below the belt, which shape the belt into a trough. Use good-quality skirting along the belt conveyor to prevent spillage. Avoid using old belt scraps to make the skirting because they can wear your conveyor belt.
Excessive Top Cover Wear, Grooving, Gouges, Rips, Ruptures, And Tears Or Stripping Of Top Cover	• **Skirt boards improperly adjusted or of wrong material.** Adjust skirt board supports to minimum 1" between metal and belt with gap increasing in direction of belt travel; use skirt board rubber (not old belt). • **Relative loading velocity too high or too low.** Adjust chutes or belt speed. Consider use of impact idlers. Observe operation closely. • **Load jams in chute.** Redesign chute for proper angle and width. Feed should be in direction of belt travel and at belt speed, centered on the belt. Control flow with feeders, chutes and skirtboards. • **Material hanging up in or under chute.** Improve loading to reduce spillage; install baffles; widen chute. • **Impact of material on belt.** Reduce impact by improving chute design; install impact idlers, or impact bed. • **Sharp edges of material or tramp iron coming in contact with cover.** Use jingle bars; impact idlers; magnetic removal equipment. • **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew: use belt designed for specific condition.** For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun. • **Breaker strip missing or inadequate.** When service is lost, install belt with proper breaker strip. • **Sticking or frozen idlers.** Free idlers and improve maintenance and lubrication.

Type Of Failure	Probable Cause / Corrective Action
Carcass Fatigue At Idler	• **Improper transition between troughed belt and terminal pulleys.** Adjust transition. • **Severe convex (hump) vertical curve.** Decrease idler spacing in curve, increase curve radius. • **Excessive forward tilt of trough rolls.** Reduce forward tilt of idlers to no more than 2° from vertical. • **Excess gap between idlers rolls.** Replace with heavier belt. • **Insufficient transverse stiffness.** Replace with the proper belt. • **Excessive sag between idlers causing load to work and shuffle on belt as it passes over idlers.** Increase tension if unnecessarily low; reduce idler spacing.
Fabric Decay, Carcass Cracks, Gouges, Ruptures, Soft Spots	• **Excessive impact of material on belt or fasteners.** Use correctly designed chutes and baffles. Make vulcanized splices. Install impact idlers. Where possible, load fines first. Where material is trapped under skirts, adjust skirtboards to minimum clearance or install cushioning idlers to hold belt against skirts. • **Material falling between belt and pulley.** Use skirtboards properly. Remove accumulation. • **Breaker strip missing or inadequate.** When service is lost, install belt with proper breaker strip. • **Drive underbelted.** Recalculate maximum belt tensions and select correct belt. If the line is over-extended, consider using two flight system with transfer point. If carcass is not rigid enough for load, install belt with proper flexibility when service is lost. • **Damage by acids, oils, chemicals, or deterioration by heat, abrasives or mildew.** Use belt designed for specific condition. For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun. Do not over lubricate idlers.
Belt Cupping - Old Belt (Was Ok When New)	• **Spilled oil or grease.** Over-lubrication of idlers; improve housekeeping; reduce quantity of grease used; check grease seals. • **Heat or chemical damage.** Use belt designed for specific condition. • **Severe pulley crowning**. Replace pulley. • **Poor belt construction**. Replace belt. • **Belt too Elastic (excessive tension)**. Replace belt.
Transverse Breaks At Belt Edge	• **Belt edges folding up on structure.** Idlers or pulleys out-of square with center line of belt: readjust idlers in affected area. Conveyor frame or structure crooked: straighten in affected area. Idler stands not centered on belt: readjust idlers in affected area. Install limit switches; provide more clearance. • **Improper transition between troughed belt and terminal pulleys.** Adjust transition in accordance with belt manufacturers recommendations. • **Severe convex (hump) vertical curve.** Decrease idler spacing in curve; increase curve radius.
Excessive Edge Wear, Broken Edges	• **Off-center loading or poor loading.** Adjust chute to place load on center of belt; discharge material in direction of belt travel at or near belt speed. Control flow with feeders, chutes and skirtboards. • **Belt strained (or elongated on one side).** Allow enough time for new belt to “break in”. If belt does not break in properly or is not new, remove strained section and splice in new piece. • **Bowed belt.** For new belt this condition should disappear during break-in; in rare instances belt must be straightened or replaced; check storage and handling of belt rolls. • **Damage by acids, chemicals, oils, or deterioration by abrasives, heat or mildew.** Use belt designed for specific condition. For abrasive materials working into cuts and between plies, repair with cold pack or with permanent repair patch. Seal metal fasteners or replace with vulcanized step splice. Enclose belt line for protection against rain, snow, or sun. Do not over lubricate idlers. • **Material spillage and build-up.** Improve loading and transfer conditions; install cleaning devices; improve maintenance. • **Belt improperly spliced.** Resplice using proper method as recommended by belt manufacturer. • **Belt hitting structure.** Install training idlers on carrying and return run.

Type Of Failure	Probable Cause / Corrective Action
Pulley / Drive Problems	
Severe Pulley Cover Wear	• **Sticking idlers.** Free idlers and improve maintenance and lubrication. • **Slippage on drive pulley.** Increase tension through screw take-up or add counterweight; lag drive pulley; increase arc of contact. • **Material spillage and build-up.** Improve loading and transfer conditions; install cleaning devices; improve maintenance. • **Material trapped between belt and pulley.** Install plows or scrapers on return run ahead of tail pulley. • **Bolt heads protruding above lagging.** Tighten bolts; replace lagging; use vulcanized-on lagging. • **Excessive forward tilt of trough rolls.** Reduce forward tilt of idlers to no more than 2° from vertical.
Tracking Problem	
Belt Runs To One Side Throughout The Entire Length At Specific Idler	• **Idlers or pulley shaft out of square with center line of conveyor.** Realign and install limit switches for greater safety. Check conveyor manufacturer's manual or guide. • **Improperly placed idlers.** Relocate idlers or insert additional idlers spaced to support belt. • **Material build-up (on pulley and idlers).** Remove accumulation and install cleaning devices, scrapers, and inverted "V" decking.
Erratic Tracking - Belt Runs Off-Line At Intermittent Points	• **Off-center loading.** Adjust chute and loading conditions so as to place load in the center of belt. • **Idlers/pulleys misaligned.** • **Insufficient pulley crowning.** • **Material buildup.** • **Belt is too stiff to train.** Use self-aligning idlers. Increase tension/conforms to crowns. Use more flexible belt on replacement. • **Tilt troughing idlers forward, but not over 2 degrees.** Use more troughable belt.
Belt Runs Off At Tail Pulley	• **Counterweight too light.** Recalculate weight required and adjust or add to counterweight or screw takeup accordingly. • **Belt running off-center around the tail pulley and through the loading area.** Install training idlers on the return run prior to tail pulley. • **Material spillage and build-up.** Improve loading and transfer conditions; install cleaning devices; improve maintenance. • **Idlers or pulleys out-of square with center line of belt.** Readjust idlers in affected area. • **Pulley lagging worn.** Replace worn pulley lagging. Use grooved lagging for wet conditions.
Belt Slips On Starting	• **Insufficient traction between belt and pulley.** Lag drive pulley; increase belt wrap; install belt cleaning devices. • **Counterweight too light.** Add counterweight or increase screw take-up tension to value determined from calculations. • **Pulley lagging worn.** Replace pulley lagging. • **Pulleys too small.** Use larger diameter pulleys. • **Improper initial positioning of counterweight in its carriage causing apparent excessive belt stretch.** Check for recommended initial position. • **Insufficient counterweight travel.** Check for recommended minimum distances.
Belt Mistracks At Head Pulley	• **Idlers/pulleys misaligned.** Realign and install limit switches for greater safety. Check conveyor manufacturer's manual or guide. • **Pulley lagging worn.** Replace worn pulley lagging. Use grooved lagging for wet conditions. Repair loose bolts protruding. • **Material buildup (on pulleys and idlers).** Remove accumulation and install cleaning devices, scrapers, and inverted "V" decking. • **Improperly placed idlers.** Relocate idlers or insert additional idlers spaced to support belt. • **Off-center loading.**

TROUBLESHOOTING

Type Of Failure	Probable Cause / Corrective Action
Belt Runs Off At All Points Of The Line	• **Skirts incorrectly placed.** Install skirtboards so that they do not rub against belt. • **Improper Loading.** Feed should be in direction of belt travel and at belt speed, centered on the belt. Control flow with feeders, chutes and skirtboards. • **Idlers or pulley shaft out of square with center line of conveyor.** Realign and install limit switches for greater safety. Check conveyor manufacturer's manual or guide. • **Material build-up (on pulleys and idlers).** Remove accumulation and install cleaning devices, scrapers, and inverted "V" decking. • **Belt strained (or elongated on one side).** Allow enough time for new belt to "break in". If belt does not break in properly or is not new, remove strained section and splice in new piece. • **Improperly placed idlers.** Relocate idlers or insert additional idlers spaced to support belt.
Belt Mistracks At Tail Pulley	• **Insufficient belt tension.** • **Idlers/pulley misaligned.** Realign and install limit switches for greater safety. Check conveyor manufacturer's manual or guide. **Idlers seized.** Free the idlers. • **Material build-up (on pulley and idlers).** Remove accumulation and install cleaning devices, scrapers, and inverted "V" decking. • **Insufficient pulley crowning.** • **Counterweight too light:** recalculate weight and adjust counter weight or screw take-up accordingly. • **Improper Loading.** Feed should be in direction of belt travel and at belt speed, centered on the belt. Control flow with feeders, chutes and skirtboards.
One Section Runs Off Line At All Conveyor Points	• **Pulleys not parallel, level, square.** • **Insufficient pulley crowning.** • **Damage by chemicals, heat, mechanical components.** • **Belt camber.** Avoid telescoping belt rolls or storing them in damp locations. A new belt should straighten out when "broken in" or complete system must be reinspected. • **Improper splice procedure/technique.** Square ends/resplice. Use fasteners recommended by the manufacturer. Retighten after running for a short while. If improperly spliced, remove old splice and resplice. Set up regular inspection schedule. • **Side Loading.** Load in direction of belt travel. • **Edge worn or Broken (permitting moisture penetration and belt shrinkage on one side).** Repair belt edge. Remove badly worn or out-of-square section and splice in new piece of belt.
Belt Runs True When Empty, Crooked When Loaded	• **Off-center loading or poor loading.** Adjust chute to place load on center of belt; discharge material in direction of belt travel at or near belt speed. • **Variations in nature and formation of load.** Use notched chute to keep load peak in exact center of belt. • **Belt not making good contact with all idlers.** Adjust height so all idlers contact belt.
Belt Runs To One Side At Given Point On Structure	• **Build-up of material on idlers.** Remove accumulation; improve maintenance. Install scrapers or other cleaning devices. • **Sticking idlers.** Free idlers and improve maintenance and lubrication. • **Idlers or pulleys out-of square with center line of belt.** Readjust idlers in affected area. • **Conveyor frame or structure crooked.** Straighten in affected area. • **Idler stands not centered on belt.** Readjust idlers in affected area. • **Structure not level.** Level structure in affected area.

Type Of Failure	Probable Cause / Corrective Action
Belt Mistracks And Runs Off At The Head Pulley	• **Spillage and worn lagging can cause material to build-up between the head pulley and belt.** Prevent the spillage and buildup by installing a slider-bedskirt board system at the loading point and a belt scraper at the head pulley. If your conveying conditions are wet and sticky, use grooved lagging on the head pulley; the grooves repel water and help prevent buildup on the belt. Improve maintenance. • **Idlers or pulleys out-of square with center line of belt.** Readjust idlers in affected area. • **Idler stands not centered on belt.** Readjust idlers in affected area.
Belts Runs To One Side Throughout The Entire Length At Specific Idler	• **Idlers or pulley shaft out of square with center line of conveyor.** Realign and install limit switches for greater safety. Check conveyor manufacturer's manual or guide. • **Improperly placed idlers.** Relocate idlers or insert additional idlers spaced to support belt. • **Material build-up (on pulley and idlers).** Remove accumulation and install cleaning devices, scrapers, and inverted "V" decking.
Belt Slips	• **Insufficient traction between belt and pulley.** Lag drive pulley - in wet conditions use grooved lagging. Increase belt wrap with snub pulleys; install belt cleaning devices and centrifugal switch for safety. • **Pulley lagging worn.** Replace pulley lagging. Ceramic lagging provides an excellent solution when conventional rubber lagging fails to correct belt slippage and premature wear. • **Counterweight too light.** Add counterweight or increase screw take-up tension to value determined from calculations. • **Material spillage and build-up.** Improve loading and transfer conditions; install cleaning devices; improve maintenance. • **Sticking or frozen idlers.** Free idlers and improve maintenance and lubrication. • **Pulleys too small.** Use larger diameter pulleys.
All Portions Of Conveyor Belt Running To One Side At A Given Point On Structure	• **One or more idlers immediately preceding trouble point not at right angles to the direction of belt travel.** Advance, in the direction of belt travel, the end of the idler to which the belt has shifted. Square idlers. • **Conveyor frame or structure crooked.** Stretch string along edge to determine extent and make correction. • **One or more idler stands not centered under belt.** Center them. Same as above. • **Sticking idlers.** Clean and lubricate. • **Belt runs off terminal pulley.** Check terminal pulley assignment. Check alignments of idlers approaching terminal pulley. • **Build-up of material on idlers.** Clean them. Install cleaning device. • **Structure not level and belt tends to shift to low side.** Level structure.
Particular Section Of Belt Runs To One Side At All Points Of Conveyor	• **Belt not joined squarely.** Remove affected splice and resplice. • **Bowed belt.** For new belt this condition should disappear during break-in; in rare instances belt must be straightened or replaced; check storage and handling of belt rolls. • **Worn edge.** "Press edge".
Belt Runs To One Side For Long Distance Or Entire Length Of Conveyor	• **Belt running off-center around the tail pulley and through the loading area.** Install training idlers on the return run prior to tail pulley. • **Off-center loading or poor loading, the result of buildup in your chutes**. Install non-stick, wear-resistant (rubber for example) chute liners to center and evenly distribute the load on the belt. Choose the liner material carefully to ensure it can resist wear from contacting your conveyed material. Adjust chute to place load on center of belt; discharge material in direction of belt travel at or near belt speed. • **Build-up of material on idlers.** Remove accumulation; improve maintenance. Install scrapers or other cleaning devices. • **Idlers or pulleys out-of square with center line of belt.** Readjust idlers in affected area. • **Conveyor frame or structure crooked.** Straighten in affected area. • **Idler stands not centered on belt.** Readjust idlers in affected area.